

# Customer Shopping Behavior Analysis

## 1. Project Overview

This Project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

## 2. Dataset Summery

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

## 3. Exploratory Data Analysis using Python

We begin with data preparation and cleaning with python

- **Data Loading:** imported the dataset using **Pandas**
- **Initial Exploration:** Used **df.info()** to check structure and **df.describe()** for summery statistics.

[18]:

| Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied | Promo Code Used | Previous Purchases | Payment Method | Frequency of Purchases |
|--------|----------------|----------|-----------------------|----------|------|-------|--------|---------------|---------------------|---------------|------------------|-----------------|--------------------|----------------|------------------------|
| 3900   | 3900           | 3900     | 3900.000000           | 3900     | 3900 | 3900  | 3900   | 3863.000000   | 3900                | 3900          | 3900             | 3900            | 3900.000000        | 3900           | 3900                   |
| 2      | 25             | 4        | NaN                   | 50       | 4    | 25    | 4      | NaN           | 2                   | 6             | 2                | 2               | NaN                | 6              | 7                      |
| Male   | Blouse         | Clothing | NaN                   | Montana  | M    | Olive | Spring | NaN           | No                  | Free Shipping | No               | No              | NaN                | PayPal         | Every 3 Months         |
| 2652   | 171            | 1737     | NaN                   | 96       | 1755 | 177   | 999    | NaN           | 2847                | 675           | 2223             | 2223            | NaN                | 677            | 584                    |
| NaN    | NaN            | NaN      | 59.764359             | NaN      | NaN  | NaN   | NaN    | 3.750065      | NaN                 | NaN           | NaN              | NaN             | 25.351538          | NaN            | NaN                    |
| NaN    | NaN            | NaN      | 23.685392             | NaN      | NaN  | NaN   | NaN    | 0.716983      | NaN                 | NaN           | NaN              | NaN             | 14.447125          | NaN            | NaN                    |
| NaN    | NaN            | NaN      | 20.000000             | NaN      | NaN  | NaN   | NaN    | 2.500000      | NaN                 | NaN           | NaN              | NaN             | 1.000000           | NaN            | NaN                    |
| NaN    | NaN            | NaN      | 39.000000             | NaN      | NaN  | NaN   | NaN    | 3.100000      | NaN                 | NaN           | NaN              | NaN             | 13.000000          | NaN            | NaN                    |
| NaN    | NaN            | NaN      | 60.000000             | NaN      | NaN  | NaN   | NaN    | 3.800000      | NaN                 | NaN           | NaN              | NaN             | 25.000000          | NaN            | NaN                    |
| NaN    | NaN            | NaN      | 81.000000             | NaN      | NaN  | NaN   | NaN    | 4.400000      | NaN                 | NaN           | NaN              | NaN             | 38.000000          | NaN            | NaN                    |
| NaN    | NaN            | NaN      | 100.000000            | NaN      | NaN  | NaN   | NaN    | 5.000000      | NaN                 | NaN           | NaN              | NaN             | 50.000000          | NaN            | NaN                    |

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype  
---  -
0   Customer ID                           3900 non-null   int64  
1   Age                                    3900 non-null   int64  
2   Gender                                3900 non-null   object  
3   Item Purchased                         3900 non-null   object  
4   Category                               3900 non-null   object  
5   Purchase Amount (USD)                 3900 non-null   int64  
6   Location                               3900 non-null   object  
7   Size                                    3900 non-null   object  
8   Color                                  3900 non-null   object  
9   Season                                 3900 non-null   object  
10  Review Rating                          3863 non-null   float64 
11  Subscription Status                    3900 non-null   object  
12  Shipping Type                          3900 non-null   object  
13  Discount Applied                       3900 non-null   object  
14  Promo Code Used                        3900 non-null   object  
15  Previous Purchases                     3900 non-null   int64  
16  Payment Method                         3900 non-null   object  
17  Frequency of Purchases                  3900 non-null   object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

```

- **Missing Data Handling:** Checked for null values and imputed missing values in the **Review Rating** column using the median rating of each product category.
- **Column Standardization:** Renamed columns to `snake case` for better readability and documentation.
- **Feature Engineering:**
  - Created `age\_group` column by binning customer ages.
  - Created `purchase\_frequency\_days` column from purchase data
- **Data Consistency Check:** Verified if `discount\_applied` and `promo\_code\_used` were redundant; dropped `promo\_code\_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

## 4.Data Analysis suing SQL(sql server)

We performed structured analysis in sql server to answer key bussiness questions.

1. **Revenue By gender:** Compared total revenue generated by male vs female employees.

|   | gender | toal_revenue |
|---|--------|--------------|
| 1 | Male   | 157890       |
| 2 | Female | 75191        |

2. **High Spending discount users:** Identified customers who used discount but still spent above the average purchase amount.

|    | customer_id | purchase_amount |
|----|-------------|-----------------|
| 1  | 2           | 64              |
| 2  | 3           | 73              |
| 3  | 4           | 90              |
| 4  | 7           | 85              |
| 5  | 9           | 97              |
| 6  | 12          | 68              |
| 7  | 13          | 72              |
| 8  | 16          | 81              |
| 9  | 20          | 90              |
| 10 | 22          | 62              |
| 11 | 24          | 88              |
| 12 | 29          | 94              |
| 13 | 32          | 79              |
| 14 | 33          | 67              |

3. **Top 5 products by Rating:** Found product with the highest average review rating.

|   | item_purchased | avg_review_rating |
|---|----------------|-------------------|
| 1 | Gloves         | 3.86              |
| 2 | Sandals        | 3.84              |
| 3 | Boots          | 3.82              |
| 4 | Hat            | 3.8               |
| 5 | Skirt          | 3.79              |

4. **Shipping type comparison:** compare average purchase amount between standard and express shipping.

|   | shipping_type | avg_purchase_amount |
|---|---------------|---------------------|
| 1 | Standard      | 58                  |
| 2 | Express       | 60                  |

5. **Subscriber Vs Non Subscribers:** Compared average spend and total revenue across subscription status.

|   | subscription_status | total_customer | avg_purchase_amount | total_purchase_amount |
|---|---------------------|----------------|---------------------|-----------------------|
| 1 | Yes                 | 1053           | 59                  | 62645                 |
| 2 | No                  | 2847           | 59                  | 170436                |

6. **Discount Depended products:** Identified 5 products with the highest percentage of discounted purchases.

|   | item_purchased | percent_Purchase |
|---|----------------|------------------|
| 1 | Hat            | 50               |
| 2 | Coat           | 49               |
| 3 | Sneakers       | 49               |
| 4 | Sweater        | 48               |
| 5 | Pants          | 47               |

7. **Customer Segmentation:** classified customer into New, Returning, and loyal segments based on purchases history.

|   | customer_segment | customer_count |
|---|------------------|----------------|
| 1 | Loyal            | 3116           |
| 2 | Returning        | 701            |
| 3 | New              | 83             |

8. **Most purchase product in a category:** Finding out the top 3 most purchased product within each category.

|    | item_rank | category    | item_purchased | total_orders |
|----|-----------|-------------|----------------|--------------|
| 1  | 1         | Accessories | Jewelry        | 171          |
| 2  | 2         | Accessories | Belt           | 161          |
| 3  | 3         | Accessories | Sunglasses     | 161          |
| 4  | 1         | Clothing    | Blouse         | 171          |
| 5  | 2         | Clothing    | Pants          | 171          |
| 6  | 3         | Clothing    | Shirt          | 169          |
| 7  | 1         | Footwear    | Sandals        | 160          |
| 8  | 2         | Footwear    | Shoes          | 150          |
| 9  | 3         | Footwear    | Sneakers       | 145          |
| 10 | 1         | Outerwear   | Jacket         | 163          |
| 11 | 2         | Outerwear   | Coat           | 161          |

9. **Repeat buyers:** Finding out the customers who are repeat buyers(more then 5 purchases) also likely subscribes.

|   | subscription_status | repeat_buyers |
|---|---------------------|---------------|
| 1 | Yes                 | 958           |
| 2 | No                  | 2518          |

10. **Revenue By age groups:** Finding out the revenue contribution from each age group.

|   | age_group   | total_revenue |
|---|-------------|---------------|
| 1 | Young Adult | 62143         |
| 2 | Middle-aged | 59197         |
| 3 | Adult       | 55978         |
| 4 | Senior      | 55763         |

## 5. Analysing with creating customer behavior dashboard (power BI)



## 5. Conclusions

### Customer Shopping Behavior Analysis | Python, SQL Server, Power BI

- Analyzed **3,900 customer transactions** to identify spending patterns, product preferences, customer segments, discount behavior, and subscription trends.
- Cleaned and prepared transactional data using **Python/Pandas**, handled missing review ratings, standardized columns, and engineered age-group and purchase-frequency features.
- Used **SQL Server** to analyze revenue, customer segmentation, repeat buyers, discount dependency, product performance, shipping preferences, and subscriber vs. non-subscriber behavior.
- Developed an interactive **Power BI dashboard** to present customer and sales insights for data-driven business decision-making.

